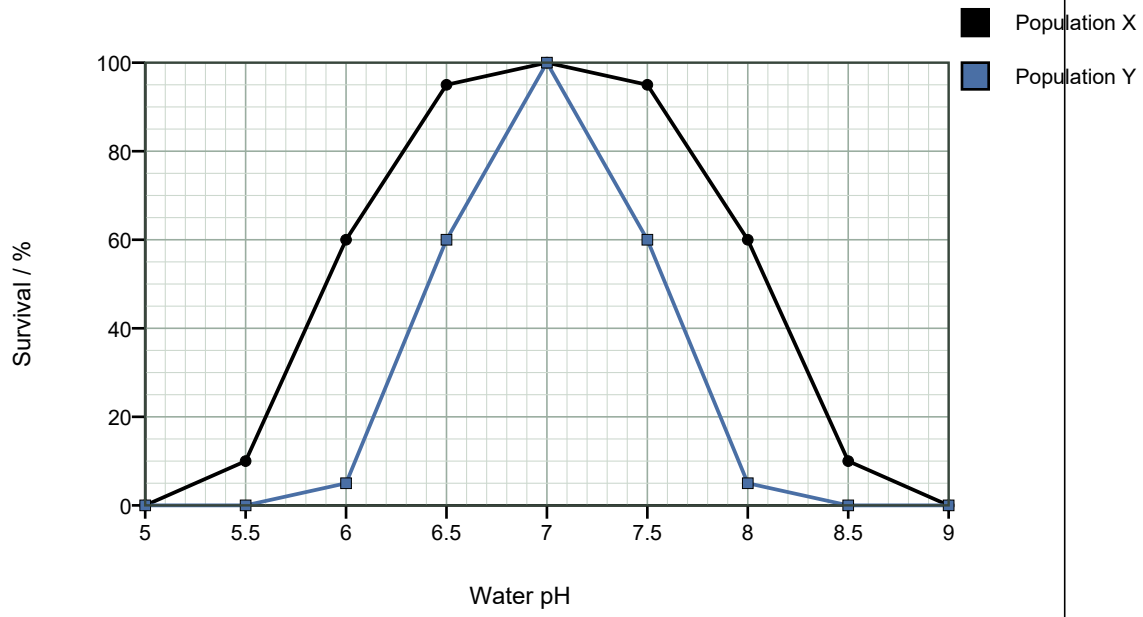
Questions by topic: Adaptation, terminology, succession & population control - Set 1 (70 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 1 . 1 Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

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0 1 . 2 A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

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0 1 . 3 Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

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0 1 . 4 Give **two** biotic factors that affect the distribution of a species.

[2 marks]

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0 2 . 1 Define each of the following ecological terms and give an example.

[8 marks]

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Species .....

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Ecological niche .....

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Community .....

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Biome .....

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0 2 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

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0 3 A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

0 3 . 1 Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

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0 3 . 2 Explain the changes shown in Table 1.

[5 marks]

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0 3 . 3 Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

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0 3 . 4 Describe the early stages of a lithosere.

[1 mark]

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0 4 . 1 Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

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0 4 . 2 Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

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0 4 . 3 Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

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0 4 . 4 Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

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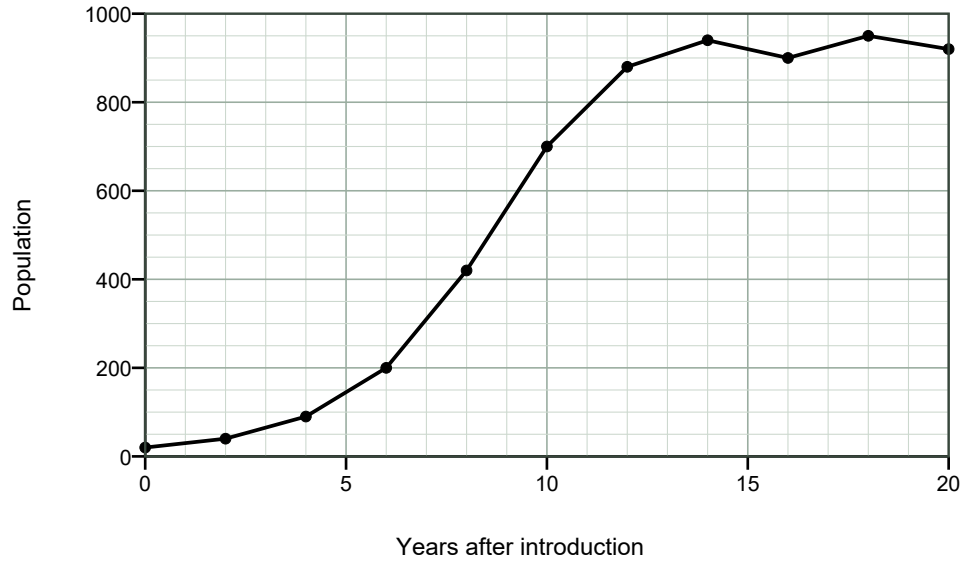
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0 5 Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 5 . 1 Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

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0 5 . 2 Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

0 5 . 3 Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

0 5 . 4 Explain how predator and prey populations regulate each other.

[2 marks]

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0 6 Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

0 6 . 1 Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

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0 6 . 2 Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

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0 6 . 3 State the equation used to forecast a change in population size.

[1 mark]

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0 6 . 4 Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

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0 7 . 1 Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

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Extra space

0 7 . 2 State what is meant by a **taxon**.

[1 mark]

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END OF QUESTIONS

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